Polar Coordinates and Conversions

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Converting points**

Convert each polar point (r, θ) to rectangular coordinates (exact values):

a $\left(4, \frac{\pi}{3}\right)$ $(2, 2\sqrt{3})$

b $\left(2, \frac{3\pi}{4}\right)$ $(-\sqrt{2}, \sqrt{2})$

(a) $(2, 2\sqrt{3})$; (b) $(-\sqrt{2}, \sqrt{2})$

2 **[Foundational]** • **Target: Converting points**

Convert each rectangular point to polar coordinates with $r \geq 0$ and $0 \leq \theta < 2\pi$:

a $(-3, 3)$ $\left(3\sqrt{2}, \frac{3\pi}{4}\right)$

b $(0, -5)$ $\left(5, \frac{3\pi}{2}\right)$

(a) $\left(3\sqrt{2}, \frac{3\pi}{4}\right)$; (b) $\left(5, \frac{3\pi}{2}\right)$

3 **[Foundational]** • **Target: Converting equations**

Convert the polar equation $r = 6\cos\theta$ to rectangular form and identify the curve.

Multiply by r: $r^2 = 6r\cos\theta$, so $x^2 + y^2 = 6x$, i.e. $(x - 3)^2 + y^2 = 9$: a circle with center $(3, 0)$ and radius 3.

4 **[Foundational]** • **Target: Converting equations**

Convert each rectangular equation to polar form:

a $x^2 + y^2 = 16$ $r = 4$

b $y = x$ $\theta = \frac{\pi}{4}$

(a) $r = 4$; (b) $\theta = \frac{\pi}{4}$

5 **[Foundational]** • **Target: Converting equations**

Convert the polar equation $r = \frac{2}{1 - \sin\theta}$ to rectangular form.

$r - r\sin\theta = 2$, so $r = y + 2$. Squaring: $x^2 + y^2 = (y + 2)^2 = y^2 + 4y + 4$, so $x^2 = 4y + 4$ — a parabola.

6 **[Foundational]** • **Target: Multiple representations**

Give two other pairs (r, θ) that represent the same point as $\left(3, \frac{\pi}{4}\right)$: one with $r > 0$ and θ outside $[0, 2\pi)$, and one with $r < 0$.

Adding a full rotation: $\left(3, \frac{\pi}{4} + 2\pi\right) = \left(3, \frac{9\pi}{4}\right)$. Using negative r (add π to the angle): $\left(-3, \frac{\pi}{4} + \pi\right) = \left(-3, \frac{5\pi}{4}\right)$.

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7 [Foundational] • Target: Multiple representations

Plot-reasoning: explain why the polar point $(-2, \frac{\pi}{2})$ corresponds to the same location as $(2, \frac{3\pi}{2})$.

A negative r means the point lies in the direction opposite to θ. Pointing at angle $\frac{\pi}{2}$ (straight up) with $r = -2$ places the point 2 units in the opposite direction, straight down — the same location as $r = 2$ at angle $\frac{3\pi}{2}$ (also straight down).

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Distance between polar points

Convert $(4, \frac{\pi}{6})$ and $(4, \frac{2\pi}{3})$ to rectangular coordinates, then find the distance between them, to two decimal places.

$(4, \frac{\pi}{6}) \to (2\sqrt{3}, 2)$; $(4, \frac{2\pi}{3}) \to (-2, 2\sqrt{3})$. Distance $= \sqrt{(2\sqrt{3}+2)^2 + (2-2\sqrt{3})^2} \approx \sqrt{29.86 + 2.14} = \sqrt{32} \approx 5.66$.

9 [Intermediate] • Target: Further conversion practice

Convert the polar equation $r^2 = 4\sin 2\theta$ to note its symmetry, and identify it as a standard curve type.

This is a lemniscate (figure-eight curve). Rewriting $r^2 = 4\sin 2\theta = 8\sin\theta\cos\theta$, and using $r^2 = x^2 + y^2$, $r\cos\theta = x$, $r\sin\theta = y$: multiplying both sides by r^2 gives $(x^2 + y^2)^2 = 8xy$.

10 [Intermediate] • Target: Why polar coordinates are not unique

Explain why, unlike rectangular coordinates, a single point can be represented by infinitely many polar coordinate pairs.

Adding any multiple of 2π to θ returns to the same direction, and reversing the sign of r while adding π to θ also gives the same point — so every point has infinitely many valid (r, θ) representations, unlike the one-to-one rectangular system.

11 [Intermediate] • Target: Why polar coordinates are not unique

The pole (origin) is a special case in polar coordinates. Explain why.

The pole corresponds to $r = 0$ for ANY value of θ — every angle gives the same point (the origin) when $r = 0$, so the pole has no single well-defined angle.

12 [Intermediate] • Target: Converting more complex equations

Convert the rectangular equation $x^2 - y^2 = 4$ (a hyperbola) to polar form.

$r^2\cos^2\theta - r^2\sin^2\theta = 4 \Rightarrow r^2(\cos^2\theta - \sin^2\theta) = 4 \Rightarrow r^2\cos(2\theta) = 4$.

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13 [Intermediate] • Target: More point conversions with negative r

Convert the polar point $(-4, \frac{\pi}{6})$ (negative r) to rectangular coordinates.

$x = r\cos\theta = -4\cos\frac{\pi}{6} = -4 \cdot \frac{\sqrt{3}}{2} = -2\sqrt{3}$. $y = r\sin\theta = -4\sin\frac{\pi}{6} = -4 \cdot \frac{1}{2} = -2$. Point: $(-2\sqrt{3}, -2)$.

14 [Intermediate] • Target: More point conversions with negative r

Convert the rectangular point $(-4, -4)$ to polar coordinates with $r > 0$ and $0 \le \theta < 2\pi$.

$r = \sqrt{16 + 16} = 4\sqrt{2}$. Both coordinates negative places this in QIII, reference angle $\frac{\pi}{4}$: $\theta = \pi + \frac{\pi}{4} = \frac{5\pi}{4}$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Converting a spiral equation • Cross-Lesson: 2.01 Right-Triangle Trigonometry

The Archimedean spiral $r = \theta$ has no simple rectangular equivalent, but explain qualitatively why it spirals outward as θ increases.

Since $r = \theta$, the radius grows in direct proportion to the angle swept — each additional full rotation (2π radians) pushes r out by 2π more units, producing evenly-spaced spiral arms rather than a closed curve.

16 [Challenging] • Target: Converting a real-world radar equation • Cross-Lesson: 2.02 Radian Measure and Arc Length

A radar station models detected objects using polar coordinates (r, θ) where r is distance in km and θ is bearing angle. An aircraft is detected at $(r, \theta) = (45, \frac{5\pi}{6})$. Convert this to rectangular coordinates (kilometers east and north of the station), to two decimal places.

$x = 45\cos\frac{5\pi}{6} = 45\left(-\frac{\sqrt{3}}{2}\right) \approx -38.97$ km. $y = 45\sin\frac{5\pi}{6} = 45\left(\frac{1}{2}\right) = 22.50$ km. The aircraft is about 38.97 km west and 22.50 km north of the station.

17 [Challenging] • Target: Converting a real-world radar equation • Cross-Lesson: 2.03 Graphs of Sine and Cosine Functions

A second aircraft is detected at rectangular coordinates $(30, -40)$ relative to the same radar station. Convert to polar coordinates (r, θ) with $r > 0$ and $0 \le \theta < 2\pi$, to two decimal places.

$r = \sqrt{30^2 + 40^2} = \sqrt{900 + 1600} = \sqrt{2500} = 50$ km. Reference angle $= \tan^{-1}\left(\frac{40}{30}\right) \approx 53.13° \approx 0.93$ rad. Since $x > 0, y < 0$ (QIV): $\theta \approx 2\pi - 0.93 \approx 5.35$ rad.

18 [Challenging] • Target: Converting a real-world radar equation • Cross-Lesson: 2.04 Law of Sines and Law of Cosines

Convert the rectangular equation of a vertical line, $x = 5$, to polar form, and explain why the result is not simply a constant r.

$r\cos\theta = 5 \Rightarrow r = \frac{5}{\cos\theta} = 5\sec\theta$. Unlike a circle centered at the pole (constant r), a vertical line's distance from the pole varies with direction θ, so r must depend on θ through the secant function.

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19 [Challenging] • Target: Converting a real-world radar equation • Cross-Lesson: 3.01 The Unit Circle and Exact Values

Convert the polar equation $r = 6$ to rectangular form, and explain in one sentence why this equation contains no θ term.

$x^2 + y^2 = 36$ (squaring both sides of $r = 6$). There is no θ term because every direction from the pole at distance 6 satisfies the equation equally — the curve is a full circle, independent of angle.

20 [Challenging] • Target: Converting a real-world radar equation • Cross-Lesson: 3.02 Fundamental Trigonometric Identities

Level 2 · MediumFind all real solutions to the trigonometric equation $2\cos^2\theta + 3\sin\theta - 3 = 0$ on the interval $[0, 2\pi)$.

Step 1 (Convert to Single Trig Ratio): Substitute $\cos^2\theta = 1 - \sin^2\theta$:

$2(1 - \sin^2\theta) + 3\sin\theta - 3 = 0 \Rightarrow 2 - 2\sin^2\theta + 3\sin\theta - 3 = 0.$ $-2\sin^2\theta + 3\sin\theta - 1 = 0 \Rightarrow 2\sin^2\theta - 3\sin\theta + 1 = 0.$ Step 2 (Factor the Quadratic in $\sin\theta$): $(2\sin\theta - 1)(\sin\theta - 1) = 0.$ Step 3 (Solve the Linear Factors): $\sin\theta = 1 \Rightarrow \theta = \frac{\pi}{2}.2\sin\theta - 1 = 0 \Rightarrow \sin\theta = \frac{1}{2}$. In $[0, 2\pi)$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Converting points

The complex number z satisfies the equation $z(2 - \mathrm{i}) = 7 + 4\mathrm{i}$. Express z in the form $x + \mathrm{i}y$, where x and y are real.

a. $2 + 3\mathrm{i}$ ✓

b. $\frac{18}{5} + \frac{1}{5}\mathrm{i}$

c. $2 - 3\mathrm{i}$

d. $\frac{10}{3} + 5\mathrm{i}$

To find z, rearrange the given equation: $z = \frac{7 + 4\mathrm{i}}{2 - \mathrm{i}}$ Multiply the numerator and the denominator by the complex conjugate of the denominator, which is $2 + \mathrm{i}$: $z = \frac{(7 + 4\mathrm{i})(2 + \mathrm{i})}{(2 - \mathrm{i})(2 + \mathrm{i})}$ Expand the denominator: $(2 - \mathrm{i})(2 + \mathrm{i}) = 2^2 - \mathrm{i}^2 = 4 - (-1) = 5$ Expand the numerator.

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22 [Foundational] • Target: Converting equations

The function f is defined for all real numbers by $f(x) = x^3 + 3x - 5$. Between which pair of consecutive integers does the unique real root of the equation $f(x) = 0$ lie?

a. 1 **and** 2 ✓

b. 0 and 1

c. 2 and 3

d. -1 and 0

To locate the root using the change of sign method, evaluate $f(x)$ at consecutive integer values: $f(0) = (0)^3 + 3(0) - 5 = -50$ Since $f(x)$ is a continuous polynomial function and there is a sign change between $x = 1$ (where $f(1)0$), by the Intermediate Value Theorem, a root of $f(x) = 0$ lies between 1 and 2 . Checking the other intervals: - On $[0, 1]$, $f(0) = -5$ and $f(1) = -1$ (no sign change). - On $[2, 3]$, $f(2) = 9$ and $f(3) = 31$ (no sign change). - On $[-1, 0]$, $f(-1) = -9$ and $f(0) = -5$.

23 [Foundational] • Target: Multiple representations

The diagram shows the graph of the function $y = f(x)$. The coordinates of the points A, B, and C are given. The graph is transformed by a translation of 3 units to the right and 2 units down. What are the new coordinates of point A?

a. **(-1, 0)** ✓

b. (5, 4)

c. (2, 0)

d. (0, -4)

The original coordinates of point A are (-4, 2). A translation of 3 units to the right means we add 3 to the x-coordinate. A translation of 2 units down means we subtract 2 from the y-coordinate. New x-coordinate = Original x-coordinate + 3 = -4 + 3 = -1 New y-coordinate = Original y-coordinate - 2 = 2 - 2 = 0 Therefore, the new coordinates of point A are (-1, 0).

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24 [Foundational] • Target: Distance between polar points

The unique real root α of the equation $x^3 + 2x - k = 0$, where k is a positive constant, is estimated using the iterative formula $x_{n+1} = \frac{k}{x_n^2 + 2}$. The sequence of iterations converges locally to α (for an initial approximation x_0 sufficiently close to α) if and only if $|g'(\alpha)| < 1$, where $g(x) = \frac{k}{x^2 + 2}$. Determine the complete set of values of k for which this iteration converges locally to α.

a. $0 < k < 4\sqrt{2}$ ✓

b. $0 < k < 2\sqrt{2}$

c. $0 < k < 3$

d. $0 < k < 12$

To determine the range of k for which the iteration converges locally, we follow these steps: 1. **Root relation:** At the fixed point $x = \alpha$, we have: $\alpha = g(\alpha) = \frac{k}{\alpha^2 + 2} \Rightarrow \alpha^3 + 2\alpha = k$. Since $k > 0$ and the function $f(x) = x^3 + 2x - k$ satisfies $f'(x) = 3x^2 + 2 > 0$ for all real x, there is a unique real root α, and $\alpha > 0$. 2. **Differentiating $g(x)$:** Using the chain rule on $g(x) = k(x^2 + 2)^{-1}$.

25 [Foundational] • Target: Further conversion practice

The complex number z is given by $z = \frac{4 + 3\mathrm{i}}{2 - \mathrm{i}}$. Express z in the Cartesian form $x + \mathrm{i}y$, where x and y are real numbers.

a. $1 + 2\mathrm{i}$ ✓

b. $1 - 2\mathrm{i}$

c. $\frac{11}{5} + \frac{2}{5}\mathrm{i}$

d. $5 + 10\mathrm{i}$

To express z in the form $x + \mathrm{i}y$, multiply the numerator and the denominator by the complex conjugate of the denominator. The denominator is $2 - \mathrm{i}$, so its complex conjugate is $2 + \mathrm{i}$. Multiply numerator and denominator: $z = \frac{(4 + 3\mathrm{i})(2 + \mathrm{i})}{(2 - \mathrm{i})(2 + \mathrm{i})}$ Expand the numerator: $(4 + 3\mathrm{i})(2 + \mathrm{i}) = 8 + 4\mathrm{i} + 6\mathrm{i} + 3\mathrm{i}^2 = 8 + 10\mathrm{i} - 3 = 5 + 10\mathrm{i}$ Expand the denominator.

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26 [Foundational] • Target: Why polar coordinates are not unique

The points A, B, and C have coordinates A(2, 5), B(k, 1), and C(8, -3). If A, B, and C are collinear, find the value of k.

a. 4 ✓

b. 5

c. 6

d. 7

For three points A, B, and C to be collinear, the gradient of the line segment AB must be equal to the gradient of the line segment BC (or AC). The gradient formula between two points (x_1, y_1) and (x_2, y_2) is given by $m = \frac{y_2 - y_1}{x_2 - x_1}$. Gradient of AB, $m_{AB} = \frac{1-5}{k-2} = \frac{-4}{k-2}$. Gradient of BC, $m_{BC} = \frac{-3-1}{8-k} = \frac{-4}{8-k}$. Equating the gradients: $\frac{-4}{k-2} = \frac{-4}{8-k}$ Since the numerators are equal and non-zero, the denominators.

27 [Foundational] • Target: Converting more complex equations

For all real values of x such that $xeq-1$ and $xeq2$, the equation $\frac{ax+7}{x+1} - \frac{b}{x-2} = \frac{2x^2-3x-20}{x^2-x-2}$ is true, where a and b are constants. What is the value of $a+b$?

a. 4

b. 6

c. 8 ✓

d. 10

To find the values of a and b, combine the fractions on the left-hand side over their common denominator, $(x+1)(x-2) = x^2 - x - 2$: $\frac{(ax+7)(x-2) - b(x+1)}{(x+1)(x-2)} = \frac{2x^2-3x-20}{x^2-x-2}$ Since the denominators are equal for all $xeq-1, 2$, their numerators must be identical for all such values of x:
$(ax+7)(x-2) - b(x+1) = 2x^2 - 3x - 20$ Expand the left side:
$ax^2 - 2ax + 7x - 14 - bx - b = 2x^2 - 3x - 20$ $ax^2 + (7 - 2a - b)x - (14 + b) = 2x^2 - 3x - 20$ Equ.

28 [Foundational] • Target: More point conversions with negative r

The function $f(x) = a \cdot 2^x + k$ has a horizontal asymptote at $y = 3$ and passes through the point $(0, 7)$, where a and k are real constants. What is the value of $f(2)$?

a. 19 ✓

b. 31

c. 16

d. 11

For an exponential function of the form $f(x) = a \cdot 2^x + k$, as $x \to -\infty$, $2^x \to 0$, which gives the horizontal asymptote $y = k$. Since the horizontal asymptote is given as $y = 3$, we have: $k = 3$ Next, the graph passes through $(0, 7)$, so $f(0) = 7$: $a \cdot 2^0 + 3 = 7$ Since $2^0 = 1$, this simplifies to: $a + 3 = 7 \Rightarrow a = 4$ Thus, the function is $f(x) = 4 \cdot 2^x + 3$. Evaluating at $x = 2$: $f(2) = 4 \cdot 2^2 + 3 = 4 \cdot 4 + 3 = 16 + 3 = 19$ Common errors: - Se.

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: Converting a spiral equation**

Consider the quadratic equation $2x^2 - (k-1)x + (k+1) = 0$ where k is a real constant. The equation has two non-real complex conjugate roots. If the imaginary parts of the roots have a magnitude of $\frac{\sqrt{7}}{2}$, what is the greatest possible value of k?

a. **7** ✓

b. 3

c. 10

d. 5

To find the parameter k, apply the quadratic formula to $ax^2 + bx + c = 0$: $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ Identify the coefficients from the given equation: $a = 2, \quad b = -(k-1), \quad c = k+1$ Compute the discriminant Δ: $\Delta = [-(k-1)]^2 - 4(2)(k+1) = (k^2 - 2k + 1) - (8k + 8) = k^2 - 10k - 7$ Since the equation has non-real complex roots, $\Delta 0$, confirming the roots are non-real. The greatest possible value of k is 7. - Misconception 3 arises from selecting the smalle.

30 **[Intermediate]** • **Target: Converting a real-world radar equation**

What are the solutions to the quadratic equation $x^2 - 6x + 13 = 0$?

a. $x = 3 \pm 2i$ ✓

b. $x = 3 \pm 4i$

c. $x = -3 \pm 2i$

d. $x = 3 \pm 2$

To solve the quadratic equation $x^2 - 6x + 13 = 0$, use the quadratic formula $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ with $a = 1$, $b = -6$, and $c = 13$: $x = \frac{-(-6) \pm \sqrt{(-6)^2 - 4(1)(13)}}{2(1)} = \frac{6 \pm \sqrt{36 - 52}}{2} = \frac{6 \pm \sqrt{-16}}{2}$ Since $\sqrt{-16} = 4i$, simplify the expression: $x = \frac{6 \pm 4i}{2} = 3 \pm 2i$ - $x = 3 \pm 4i$ occurs if the radical term is not divided by the denominator 2. - $x = -3 \pm 2i$ occurs from a sign error when computing $-b$..

31 **[Intermediate]** • **Target: Converting points**

Consider the quadratic equation $x^2 - 8x + 15 = 0$. What are the roots of this equation?

a. $x = 3$ **and** $x = 5$ ✓

b. $x = -3$ and $x = -5$

c. $x = 3$ and $x = -5$

d. $x = -3$ and $x = 5$

To find the roots of the quadratic equation $x^2 - 8x + 15 = 0$, we can factor the quadratic expression. We are looking for two numbers that multiply to 15 and add up to -8. These numbers are -3 and -5. Therefore, the factored form of the equation is $(x-3)(x-5) = 0$. Setting each factor to zero, we get $x - 3 = 0$ which gives $x = 3$, and $x - 5 = 0$ which gives $x = 5$. Thus, the roots are $x = 3$ and $x = 5$.

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32 [Intermediate] • Target: Converting equations

Which is NOT a quadratic equation?

a. $2x^2 = 5$

b. $x^2 + x + 1 = 0$

c. $x^2 - 1 = 0$

d. $x^3 + 2x = 0$ ✓

Applying Quadratics & Complex Numbers principles, Cubic, not quadratic. Therefore, the verified correct option is $x^3 + 2x = 0$.

33 [Intermediate] • Target: Multiple representations

An ellipse has equation $\frac{x^2}{36} + \frac{y^2}{20} = 1$. What are the coordinates of its foci?

a. $(\pm 5, 0)$

b. $(\pm 4, 0)$ ✓

c. $(0, \pm 4)$

d. $(\pm 6, 0)$

$c^2 = a^2 - b^2 = 36 - 20 = 16 \Rightarrow c = 4$. Since $a^2 > b^2$, foci lie along the x-axis at $(\pm 4, 0)$.

34 [Intermediate] • Target: Distance between polar points

Find the x-coordinates of the points of intersection of $y = x^2 - 9x + 22$ and $y = 2$.

a. **x=4 or x=5** ✓

b. x=-4 or x=-5

c. x=5 or x=5

d. x=4 or x=4

Substituting the line into the curve gives $x^2 - 9x + 20 = 0$, which factorises as $(x - 5)(x - 4) = 0$, so $x = 4$ or $x = 5$.

35 [Intermediate] • Target: Further conversion practice

The line $y = -12$ intersects the circle $x^2 + y^2 = 169$ at two points. What is the distance between the two intersection points?

a. **10** ✓

b. 13

c. 16

d. 5

$x^2 = 169 - (-12)^2 = 25 \Rightarrow x = \pm 5$. Distance $= 2(5) = 10$.

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36 [Intermediate] • Target: Why polar coordinates are not unique

If z and $\bar{z}$ are roots of $x^2 + bx + c = 0$ (real coefficients), express b and c in terms of $a + bi$.

a. $b = 0,\ c = a^2$

b. $b = a,\ c = b$

c. $b = -2a,\ c = a^2 + b^2$ ✓

d. $b = 2a,\ c = a^2 - b^2$

Sum of roots $= 2a = -b \Rightarrow b = -2a$. Product $= |z|^2 = a^2 + b^2 = c$.

37 [Intermediate] • Target: Converting more complex equations

Which equation has roots that are complex conjugates?

a. $x^2 - 5 = 0$

b. $x^2 + 2x - 5 = 0$

c. $x^2 - 2x + 5 = 0$ ✓

d. $x^2 - 2x - 5 = 0$

Discriminant $= 4 - 20 < 0$: complex roots.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: More point conversions with negative r • Cross-Lesson: 2.04 Law of Sines and Law of Cosines

Find the distance between the points $(7, 4)$ and $(16, -8)$.

a. 21

b. 14

c. **15** ✓

d. 16

$d = \sqrt{(9)^2 + (12)^2} = \sqrt{225} = 15$.

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39 [Challenging] • Target: Converting a spiral equation • Cross-Lesson: 3.01 The Unit Circle and Exact Values

Derive the polar equation of a conic with focus at the pole and directrix perpendicular to the polar axis at distance d.

a. $r = d/\theta$

b. $r = e/(1 - d\cos\theta)$

c. $r = ed\cos\theta$

d. $r = \frac{ed}{1 - e\cos\theta}$ **(directrix to the left)** ✓

Standard polar conic derivation.

40 [Challenging] • Target: Converting a real-world radar equation • Cross-Lesson: 3.02 Fundamental Trigonometric Identities

The graph of the exponential function $f(x) = a \cdot b^x + c$ is shown in the xy-plane, where a, b, and c are constants with $b > 0$ and $beq1$. The function has a horizontal asymptote at $y = -4$ and passes through the points $(0, -1)$ and $(2, 8)$. What is the value of $f(-1)$?

a. $-\frac{5}{2}$ ✓

b. $-\frac{7}{2}$

c. $-\frac{13}{4}$

d. $\frac{3}{2}$

To determine $f(-1)$, we first find the constants a, b, and c: 1. **Horizontal Asymptote:** For an exponential function of the form $f(x) = a \cdot b^x + c$, as the exponential term approaches zero, the function approaches its horizontal asymptote $y = c$. Given that the asymptote is $y = -4$, we have: $c = -4$ 2. **Use the Point $(0, -1)$:** Substitute $x = 0$ and $f(0) = -1$: $f(0) = a \cdot b^0 - 4 = -1$ $a(1) - 4 = -1 \Rightarrow a = 3$ 3. **Use the Point $(2, 8)$:** Substitute $x = 2$ and $f(2) = 8$.

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41 [Challenging] • Target: Converting points • Cross-Lesson: 3.03 Sum and Difference Formulas

Consider the matrix equation $(2,1,-2;4,-1,-1;2,-2,1)(x;y;z)=(9;9;k)$ where k is a real constant. Given that the system has infinitely many real solutions (x,y,z), what is the minimum possible value of $x^2+y^2+z^2$?

a. **9** ✓

b. 3

c. 6

d. 45

We solve the problem in three steps: determine k for consistency, parameterize the infinite solution set, and minimize the quadratic objective function. **Step 1: Determine k for consistency** Write the augmented matrix of the system: $(2,1,-2,9;4,-1,-1,9;2,-2,1,k)$ Perform Gaussian elimination: - Replace Row 2 with R_2-2R_1: $(4-4,\ -1-2,\ -1-(-4)\mid 9-18)=(0,-3,3\mid -9)$, which simplifies to $(0,1,-1\mid 3)$. - Replace Row 3 with R_3-R_1.

42 [Challenging] • Target: Converting equations • Cross-Lesson: 3.04 Double-Angle Formulas and Trigonometric Equations

Consider the function $f(x)=4\cos^2(x)+b\sin(x)+c$ where b and c are real constants with $b>0$. If the maximum value of $f(x)$ is 10 and the minimum value of $f(x)$ is 1, what is the value of c?

a. 5 ✓

b. $\frac{11}{2}$

c. 9

d. 4

To find the unknown parameters, we first express $f(x)$ in terms of a single trigonometric function using the identity $\cos^2(x)=1-\sin^2(x)$: $f(x)=4(1-\sin^2(x))+b\sin(x)+c=-4\sin^2(x)+b\sin(x)+c+4$ Let $u=\sin(x)$. Since $x\in\mathbb{R}$, the variable u is restricted to the interval $[-1,1]$. We analyze the quadratic function: $g(u)=-4u^2+bu+c+4,\quad u\in[-1,1]$ The graph of $g(u)$ is a downward-opening parabola with its vertex at: $u_v=-\frac{b}{2(-4)}=\frac{b}{8}$ S.

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43 [Challenging] • Target: Multiple representations • Cross-Lesson: 4.01 Circles and Parabolas

Determine the complete set of real values of the constant k for which the equation $2\cos 2\theta - 4\sin\theta = k$ has at least one real solution for θ.

a. $-6 \le k \le 3$ ✓

b. $-6 \le k \le 2$

c. $k \le 3$

d. $-3 \le k \le 6$

To find the set of values of k for which real solutions exist, we express the left-hand side entirely in terms of $\sin\theta$. Using the double-angle identity $\cos 2\theta = 1 - 2\sin^2\theta$: $2(1 - 2\sin^2\theta) - 4\sin\theta = k$ $-4\sin^2\theta - 4\sin\theta + 2 = k$ Let $u = \sin\theta$. Since $\theta \in \mathbb{R}$, u is constrained to the closed interval $[-1, 1]$. The problem reduces to finding the range of the quadratic function $f(u) = -4u^2 - 4u + 2, \quad u \in [-1, 1]$. 1. **Find the vertex (s.

44 [Challenging] • Target: Distance between polar points • Cross-Lesson: 4.02 Ellipses

Find the set of all real solutions to the equation $\log_2(x-1) - \log_4(x+14) = 1$

a. $x = 11$ **only** ✓

b. $x = 11$ and $x = -5$

c. $x = 5$ only

d. $x = 16$ only

To solve the equation, we first express all logarithms in the same base using the change of base formula: $\log_4(x+14) = \frac{\log_2(x+14)}{\log_2 4} = \frac{\log_2(x+14)}{2} = \frac{1}{2}\log_2(x+14)$ Substitute this back into the original equation: $\log_2(x-1) - \frac{1}{2}\log_2(x+14) = 1$ Multiply through by 2: $2\log_2(x-1) - \log_2(x+14) = 2$ Apply the power law and the quotient law of logarithms.

45 [Challenging] • Target: Further conversion practice • Cross-Lesson: 4.03 Hyperbolas

The rational function f is defined for all real $xotin\{-1, 2\}$ by $f(x) = \frac{x^3 + kx + 7}{(x+1)(x-2)^2}$ where k is a constant. The partial fraction decomposition of $f(x)$ takes the form $A + \frac{B}{x+1} + \frac{C}{x-2} + \frac{D}{(x-2)^2}$ where $A, B, C,$ and D are real constants. Given that $C = 2$, find the value of k.

a. -3 ✓

b. 1

c. 3

d. 24

Step 1: Determine the polynomial quotient A. The degree of the numerator is 3 and the degree of the denominator is $1 + 2 = 3$. Expanding the denominator gives: $(x+1)(x-2)^2 = (x+1)(x^2 - 4x + 4) = x^3 - 3x^2 + 4$ Since the leading coefficient of both the numerator and the denominator is 1, polynomial division yields a constant quotient: $A = \frac{1}{1} = 1$ Step 2: Set up the algebraic identity. Multiplying both sides of the partial fraction expansion by $(x+1)(x-2)^2$.

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